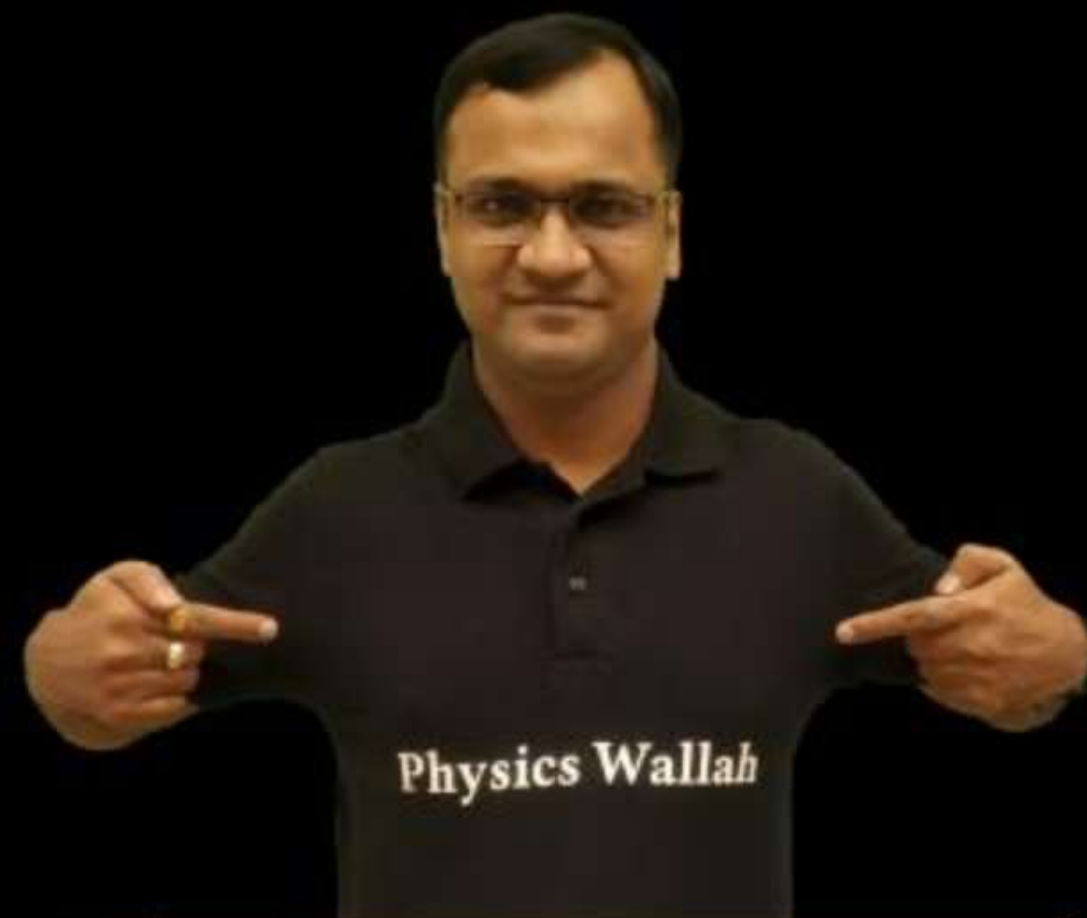
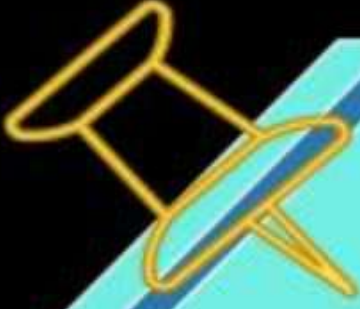


PRA_YAS 2.0

Chapter Name: Determinant
Lecture No. - 03



RAJEEV RASTOGI SIR

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Topics to be covered

1. System of Linear equation ✓

2. Cramer's rule ✓

3. Mixed problem practice ✓



Q. Which of the following values of α satisfy the equation

$$\begin{vmatrix} (1 + \alpha)^2 & (1 + 2\alpha)^2 & (1 + 3\alpha)^2 \\ (2 + \alpha)^2 & (2 + 2\alpha)^2 & (2 + 3\alpha)^2 \\ (3 + \alpha)^2 & (3 + 2\alpha)^2 & (3 + 3\alpha)^2 \end{vmatrix} = -648 \alpha ?$$

[JEE(Advanced)-2015, 4M, -2M]

A. -4

B. 9

C. -9

D. 4

Solⁿ

$$\begin{vmatrix} \underline{1+2x+x^2} \checkmark & \underline{1+4x+4x^2} \checkmark & \underline{1+6x+9x^2} \checkmark \\ \underline{9+9x+x^2} \checkmark & 9+8x+4x^2 \checkmark & 9+12x+9x^2 \checkmark \\ \underline{9+6x+x^2} \checkmark & 9+12x+4x^2 \checkmark & 9+18x+9x^2 \checkmark \end{vmatrix} = -648x$$

$$\Rightarrow \begin{vmatrix} 1 & x & x^2 \\ 4 & 2x & x^2 \\ 9 & 3x & x^2 \end{vmatrix} \times \begin{vmatrix} 1 & 2 & 1 \\ 1 & 4 & 4 \\ 1 & 6 & 9 \end{vmatrix} = -648x$$

$$x^3 \begin{vmatrix} 1 & 1 & 1 \\ 4 & 2 & 1 \\ 9 & 3 & 1 \end{vmatrix} \begin{matrix} \uparrow 12 \\ \left[1 \cdot (36 - 24) - 2(5) + 1 \cdot (2) \right] \end{matrix} = -162x$$

$$x^3 \left[1 \cdot (2-3) - 1 \cdot (9-9) + 1 \cdot (12-18) \right] = -162x$$

~~-1+5-6~~

$$x^2 (-1+5-6) = -162$$

$x^2 = 81$

$$x = 9, -9, 0$$

Cofactor and Minors of an element:

Minors: Minors of an element is defined as the minor determinant obtained by deleting a particular row or column in which that element lies. E.g. in the determinant

$$D = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \text{ minor of } a_{12} \text{ denoted as } M_{12} = \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} \text{ and so on}$$

Cofactor $\rightarrow$ Signed minor

$$M_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$$

$$C_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} = M_{11} \quad C_{12} = - \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} = -M_{12}$$

Cofactor: It has no separate identity and is related to the Minor as

$$C_{ij} = (-1)^{i+j} M_{ij} \text{ where 'i' denotes the row and 'j' denotes the column.}$$

Hence the value of a determinant of order three in terms of 'Minor' and 'Cofactor' can be written as

$$D = a_{11} M_{11} - a_{12} M_{12} + a_{13} M_{13} \text{ or} \\ = a_{11} C_{11} + a_{12} C_{12} + a_{13} C_{13}$$

$$D = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

Also, $a_{11} C_{11} + a_{12} C_{12} + a_{13} C_{13} = 0$

There are 12 such expressions. ✓

$$D = a_{11} M_{11} - a_{12} M_{12} + a_{13} M_{13}$$

$$D = a_{11} C_{11} + a_{12} C_{12} + a_{13} C_{13}$$

There are six such expressions

$$D = a_{31} C_{31} + a_{32} C_{32} + a_{33} C_{33} \checkmark$$

Consider a determinant

$$\text{expand} \rightarrow \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$= (-1)^{1+1} a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} + (-1)^{1+2} a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + (-1)^{1+3} a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

$$\Rightarrow a_{11} \cdot C_{11} + a_{12} \cdot C_{12} + a_{13} \cdot C_{13} = 0.$$

Important Result :-

If $D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ and $\Delta = \begin{vmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{vmatrix}$ where Δ is the determinant obtained by replacing each element by its cofactors, then

$$\Delta = D^2 \quad (D \neq 0)$$

Proof :-

$$D \cdot \Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \begin{vmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{vmatrix} \quad \text{MwL R by R}$$

$$= \begin{vmatrix} D & 0 & 0 \\ 0 & D & 0 \\ 0 & 0 & D \end{vmatrix} = D^3 \Rightarrow \boxed{\Delta = D^2}$$

System of Equations:-



Consistent system:- A system is said to be consistent if it has a solⁿ.
(either unique solⁿ or infinite solⁿ)

In consistent system:- ————— inconsistent if it has no solⁿ.

Trivial solution:- If all the variables involved in the system are simultaneously zero and satisfying the system, then this solⁿ is known as trivial solⁿ.

Non-Trivial Solution :- If at least one of the variable involved in the system is non-zero and satisfying the system then it is known as non-trivial solⁿ. -

Cramer's Rule: [Simultaneous Equations Involving Three Unknown]

Let $\left. \begin{aligned} a_1x + b_1y + c_1z &= d_1 \\ a_2x + b_2y + c_2z &= d_2 \\ a_3x + b_3y + c_3z &= d_3 \end{aligned} \right\} \begin{aligned} &\dots(i); \\ &\dots(ii); \\ &\dots(iii) \end{aligned}$

Then, $x = \frac{D_1}{D}, y = \frac{D_2}{D}, z = \frac{D_3}{D}$, (if $D \neq 0$)

Where $D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}; D_1 = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}; D_2 = \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix}$

& $D_3 = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}$

Proof

$$D_1 = \begin{vmatrix} \textcircled{d_1} & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1x + b_1y + c_1z & b_1 & c_1 \\ a_2x + b_2y + c_2z & b_2 & c_2 \\ a_3x + b_3y + c_3z & b_3 & c_3 \end{vmatrix} \quad c_1 \rightarrow c_1 - y \cdot c_2 - z \cdot c_3$$

$$\begin{aligned} 0 &= x \cdot 0 \\ 0 &= y \cdot 0 \\ 0 &= z \cdot 0 \end{aligned}$$

|||

Δ

$$\begin{aligned} D_1 &= x \cdot D \\ D_2 &= y \cdot D \\ D_3 &= z \cdot D \end{aligned}$$

Con-I if $D \neq 0$

then the system has unique solⁿ
given by $x = \frac{D_1}{D}, y = \frac{D_2}{D}, z = \frac{D_3}{D}$

Con-II if $D = 0$

Find D_1, D_2, D_3

ex(ceptional case)-

$$\left. \begin{aligned} x + y + z &= 1 \\ x + y + z &= 2 \\ x + y + z &= 3 \end{aligned} \right\} \begin{array}{l} \text{System of} \\ \text{3 planes} \end{array}$$

No solⁿ

If at least one of D_1, D_2, D_3 is non-zero
 $\Rightarrow$ system has no solⁿ

If $D_1 = D_2 = D_3 = 0$
 $\Rightarrow$ system has infinitely many solutions.



Q.

For what values p and q the system of equation

$$2x + py + 6z = 8$$

$$x + 2y + qz = 5$$

$$x + y + 3z = 4$$

has (i) no solution (ii) unique solution (iii) infinite solution.

$$D = \begin{vmatrix} 2 & p & 6 \\ 1 & 2 & q \\ 1 & 1 & 3 \end{vmatrix}$$

$$= 2(6 - q) - p(3 - q) + 6(1 - 2)$$

$$= 12 - 2q - 3p + pq - 6$$

$$= 6 - 2q - 3p + pq$$

$$= 2(3 - q) - p(3 - q)$$

$$\Rightarrow D = (p - 2)(q - 3)$$

$$D_1 = \begin{vmatrix} 8 & p & 6 \\ 5 & 2 & q \\ 4 & 1 & 3 \end{vmatrix}$$

$$= 8(6 - q) - p(15 - 4q) + 6(5 - 8)$$

$$= 48 - 8q - 15p + 4pq - 18$$

$$= 30 - 8q - 15p + 4pq$$

$$= 15(2 - p) - 4q(2 - p)$$

$$D_1 = (p - 2)(4q - 15)$$

$$D_3 = \begin{vmatrix} 2 & p & 8 \\ 1 & 2 & 5 \\ 1 & 1 & 4 \end{vmatrix}$$

$$= 2(8 - 5) - p(4 - 5) + 8(1 - 2)$$

$$= 6 + p - 8$$

$$D_3 = p - 2$$



Q.

$$\text{Prove that } 3x - y + 4z = 3 \quad (1)$$

$$x + 2y - 3z = -2 \quad (2)$$

$$6x + 5y + \lambda z = -3 \quad (3)$$

has at least one solution for any real λ .

Find set of solution when $\lambda = -5$.

$$D_2 = \begin{vmatrix} 3 & 3 & 4 \\ 1 & -2 & -3 \\ 6 & -3 & 1 \end{vmatrix}$$

$$= 3(-2 \cdot 1 - 9) - 3(1 + 18) + 4(-3 + 12) = -6 - 27 - 3 - 54 + 36 = -91 - 45 = -136$$

$$D = \begin{vmatrix} 3 & -1 & 4 \\ 1 & 2 & -3 \\ 6 & 5 & 1 \end{vmatrix}$$

$$= 3(2 \cdot 1 + 15) + 1 \cdot (1 + 18) + 4(5 - 12)$$

$$= 6 \cdot 1 + 19 + 4(-7)$$

$$= 6 + 19 - 28 = -7 \Rightarrow D = -7(1 + 5)$$

$$D_1 = \begin{vmatrix} 3 & -1 & 4 \\ -2 & 2 & -3 \\ -3 & 5 & 1 \end{vmatrix}$$

$$= 3(2 \cdot 1 + 15) + 1 \cdot (-2 \cdot 1 - 9)$$

$$+ 4(-10 + 6)$$

$$= 6 + 19 - 28 - 9 - 16 = -18$$

$$= -18 = -9(1 + 5)$$

$$D_3 = \begin{vmatrix} 3 & -1 & 3 \\ 1 & 2 & -2 \\ 6 & 5 & -3 \end{vmatrix}$$

$$= 3(-6 + 10) + 1 \cdot (-3 + 12)$$

$$+ 3(5 - 12)$$

$$= 12 + 9 - 21 = 0 \Rightarrow D_3 = 0$$

$$D = 7(1+5)$$

$$D_1 = 9(1+5)$$

$$D_2 = -9(1+5)$$

$$D_3 = 0$$

$$\text{if } 1+5 \neq 0 \Rightarrow 1 \neq -5 \Rightarrow D \neq 0$$

$$\left. \begin{aligned} X &= \frac{D_1}{D} = \frac{9}{7} \\ Y &= \frac{D_2}{D} = -\frac{9}{7} \\ Z &= \frac{D_3}{D} = 0 \end{aligned} \right\}$$

$$(X, Y, Z) = \left(\frac{9-5k}{9}, \frac{13k-9}{7}, k \right) \quad \text{if } k \neq R$$

$$\text{if } 1 = -5$$

$\Downarrow$

$$D = D_1 = D_2 = D_3 = 0$$

$\Downarrow$ system has infinite soln.

$$\text{let } z = k$$

$$\text{then (1)} \Rightarrow 3X - Y = 3 - 4k$$

$$X + 2Y = -2 + 3k$$

$$7X = 4 - 5k$$

$$X = \frac{4-5k}{7}$$

$$2Y = -2 + 3k - \left(\frac{4-5k}{7} \right)$$

$$Y = \frac{-14 + 21k - 4 + 5k}{14} = \frac{-18 + 26k}{14} = \frac{13k-9}{7}$$

Homogeneous system of equation :-

$$a_1x + b_1y + c_1z = 0$$

$$a_2x + b_2y + c_2z = 0$$

$$a_3x + b_3y + c_3z = 0$$

Homogeneous system is always consistent
beoz trivial solⁿ always exists.

OKAY

$$\begin{matrix} x=0 \\ y=0 \\ z=0 \end{matrix}$$

Note: Trivial solution $(0, 0, 0)$ always exists and for non-trivial solution $D = 0$ ✓

$$\begin{matrix} 0 = x \cdot D \checkmark \\ 0 = y \cdot D \checkmark \\ 0 = z \cdot D \checkmark \end{matrix}$$

$$\begin{matrix} D_1 = x \cdot D \\ D_2 = y \cdot D \\ D_3 = z \cdot D \end{matrix}$$

$$\text{here } D_1 = D_2 = D_3 = 0.$$



Q. The number of values of $\theta \in (0, \pi)$ for which the system of linear equations

$$x + 3y + 7z = 0$$

$$-x + 4y + 7z = 0$$

$$(\sin 3\theta)x + (\cos 2\theta)y + 2z = 0$$

has a non-trivial solution, is:

(cond is $D=0$)

$$\Rightarrow \begin{vmatrix} 1 & 3 & 7 \\ -1 & 4 & 7 \\ \sin 3\theta & \cos 2\theta & 2 \end{vmatrix} = 0$$

$$\sin 3\theta (21 - 28) - \cos 2\theta (7 + 7) + 2(4 + 3) = 0$$

[JEE (Main)-2019] ✓

A. three

B. two

C. four

D. one

$$\begin{aligned} \sin 3\theta + 2 \cos 2\theta - 2 &= 0 \\ 3 \sin \theta - 4 \sin^3 \theta + 2(1 - 2 \sin^2 \theta) - 2 &= 0 \end{aligned}$$

$$\begin{aligned} -7 \sin 3\theta - 14 \cos 2\theta + 14 &= 0 \\ \rightarrow 3s - 4s^3 - 4s^2 &= 0 \\ s(4s^2 + 4s - 3) &= 0 \end{aligned}$$

$$s(4s^2 + 6s - 2s - 3) = 0$$

$$s(2s(2s+3) - 1 \cdot (2s+3)) = 0$$

$$\underbrace{s}_{\neq 0} \underbrace{(2s+3)}_{\neq 0} (2s-1) = 0$$

$$\downarrow \sin \theta = \frac{1}{2}$$

$$\boxed{\theta = 30^\circ, 150^\circ}$$



Q. The system of linear equations

$$x + y + z = 2$$

$$2x + 3y + 2z = 5$$

$$2x + 3y + (a^2 - 1)z = a + 1$$

$$D = \begin{vmatrix} 1 & 1 & 1 \\ 2 & 3 & 2 \\ 2 & 3 & (a^2 - 1) \end{vmatrix} \quad R_3 \rightarrow R_3 - R_2$$

[JEE(Main)-2019]

A. is inconsistent when $a = 4$ (X)

B. has a unique solution for $|a| = \sqrt{3}$ X

C. has infinitely many solutions for $a = 4$ X

D. is inconsistent when $|a| = \sqrt{3}$



$$2x + 3y + 2z = \sqrt{3} + 1$$

$$= \begin{vmatrix} 1 & 1 & 1 \\ 2 & 3 & 2 \\ 0 & 0 & a^2 - 3 \end{vmatrix}$$

$$D = (a^2 - 3) \cdot 1$$

$$\text{if } |a| = \sqrt{3} \quad D = 0 \quad \text{Not } 0$$



Q. The set of all values of λ for which the system of linear equations

$$x - 2y - 2z = \lambda x \qquad x + 2y + z = \lambda y \qquad -x - y = \lambda z$$

has a non-trivial solution: [JEE mains-2020]

- A. is a singleton
- B. contains exactly two elements
- C. is an empty set
- D. contains more than two elements

$$(1-\lambda)x - 2y - 2z = 0$$

$$x + (2-\lambda)y + z = 0$$

$$-x - y + \lambda z = 0$$

Cond. for non-trivial solⁿ

$$D=0 \Rightarrow \begin{vmatrix} 1-\lambda & -2 & -2 \\ 1 & 2-\lambda & 1 \\ -1 & -1 & \lambda \end{vmatrix} = 0$$

No. of solⁿ



Q.

If the system of linear equations

$$x + y + 3z = 0$$

$$3x + y + 3z = 0$$

$$x + 3y + k^2z = 0$$

$$D=0$$
$$k=-3$$

has a non-zero solution (x, y, z) for some $k \in \mathbb{R}$ then

$x + \left(\frac{y}{z}\right)$ is equal to

[JEE main - 2020]

A. 9

B. 3

C. -9

D. -3

$$x=0, y=-3z$$
$$x + \frac{y}{z} = 0 - 3 = -3 \checkmark$$

$$\frac{x}{3-3} = \frac{y}{9-3} = \frac{z}{1-3}$$
$$\Rightarrow \frac{x}{0} = \frac{y}{6} = \frac{z}{-2} = \lambda$$
$$x=0, y=6\lambda, z=-2\lambda$$



Q. The number of values of k for which the system of equation
 $(k+1)x + 8y = 4k$ $kx + (k+3)y = 3k-1$
has infinitely many solutions is [JEE 2002 (Screening), 3]

A. 0

B. 1

C. 2

D. infinite

$$\left. \begin{array}{l} x+y=4 \\ 2x+2y=8 \end{array} \right\}$$

$$\frac{12}{8} = \frac{3}{2}$$

Condition is

$$\frac{k+1}{k} = \frac{8}{k+3} = \frac{4k}{3k-1}$$

$$(k+1)(k+3) = 8k$$
$$k^2 - 4k + 3 = 0$$

$k=1, 3$ not a solⁿ

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

$$a_1x + b_1y = c_1$$

$$a_2x + b_2y = c_2$$

Unique solⁿ

No solⁿ

Infinitely many solⁿ

$$-\frac{a_1}{b_1} \neq -\frac{a_2}{b_2} \Rightarrow \frac{a_1}{a_2} \neq \frac{b_1}{b_2}$$

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$



Q. If the equations $a(y + z) = x$, $b(z + x) = y$, $c(x + y) = z$ have nontrivial solutions, then find the value of $\frac{1}{1+a} + \frac{1}{1+b} + \frac{1}{1+c}$.

$$x - ay - az = 0$$

$$bx - y + bz = 0$$

$$cx + cy - z = 0$$

$$\text{Cond. is } D = 0$$

$$\begin{vmatrix} 1 & -a & -a \\ b & -1 & b \\ c & c & -1 \end{vmatrix} = 0$$

$$\begin{aligned} & (1 - a - a) \text{ expand} \\ & (2 - 2 - 1) \end{aligned}$$

$$E = \frac{1}{1 + \frac{x}{y+z}} + \frac{1}{1 + \frac{y}{z+x}} + \frac{1}{1 + \frac{z}{x+y}}$$

$$E = \frac{y+z + z+x + x+y}{x+y+z}$$

$$E = 2$$



Q.

Solve the system of equation ;

$$z + ay + a^2x + a^3 = 0$$

$$z + by + b^2x + b^3 = 0$$

$$z + cy + c^2x + c^3 = 0$$

$$D = \begin{vmatrix} a^2 & a & 1 \\ b^2 & b & 1 \\ c^2 & c & 1 \end{vmatrix} = -(a-b)(b-c)(c-a)$$

$$D_1 = \begin{vmatrix} -a^3 & a & 1 \\ -b^3 & b & 1 \\ -c^3 & c & 1 \end{vmatrix}$$

$$D_2 = \begin{vmatrix} & & \\ & & \\ & & \end{vmatrix}$$

$$D_3 = \begin{vmatrix} & & \\ & & \\ & & \end{vmatrix}$$

Cool Approach:-

Consider an eqn $t^3 + t^2x + ty + z = 0$ $\begin{matrix} a \\ b \\ c \end{matrix}$

$$a + b + c = -x$$

$$ab + bc + ca = y$$

$$abc = -z$$



Q. If $ax_1^2 + by_1^2 + cz_1^2 = ax_2^2 + by_2^2 + cz_2^2 = ax_3^2 + by_3^2 + cz_3^2 = d$
and $ax_2x_3 + by_2y_3 + cz_2z_3 = ax_3x_1 + by_3y_1 + cz_3z_1 = ax_1x_2 + by_1y_2$
 $+ cz_1z_2 = f$, then prove that $\begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} = (d - f) \left[\frac{d+2f}{abc} \right]^{1/2}$
(a, b, c $\neq$ 0)

(11.11)

Question of the day



- Q. Let S be the set of all integer solutions, (x, y, z) , of the system of equations
- $$\begin{aligned}x - 2y + 5z &= 0 & -2x + 4y + z &= 0 \\ -7x + 14y + 9z &= 0\end{aligned}$$
- such that $15 \leq x^2 + y^2 + z^2 \leq 150$. Then, the number of elements in the set S is equal to [JEE main-2020] ✓✓



Q. The total number of distinct $x \in \mathbb{R}$ for which

$$\begin{vmatrix} x & x^2 & 1+x^3 \\ 2x & 4x^2 & 1+8x^3 \\ 3x & 9x^2 & 1+27x^3 \end{vmatrix} = 10 \text{ is}$$

hw.

[JEE (Advanced)-2016, 3(0)]



Q. Let $a, \lambda, m \in \mathbb{R}$. Consider the system of linear equations
 $ax + 2y = \lambda$ $3x - 2y = \mu$. Which of the following h.w
statement(s) is/are correct? [JEE Adv. 2016, 4(-2)]

A. If $a = -3$, then the system has infinitely many solutions for all value of λ and μ

B. If $a \neq -3$, then the system has a unique solution for all values of λ and μ

C. If $\lambda + \mu = 0$, then the system has infinitely many solution for $a = -3$

D. If $\lambda + \mu \neq 0$, then the system has no solutions for $a = -3$



- Q. The number of all possible values of θ , where $0 < \theta < \pi$, for which the system of equations
- $$(y + z) \cos 3\theta = (xyz) \sin 3\theta$$
- $$x \sin 3\theta = \frac{2 \cos 3\theta}{y} + \frac{2 \sin 3\theta}{z}$$
- $$(xyz) \sin 3\theta = (y + 2z) \cos 3\theta + y \sin 3\theta$$
- have a solution (x_0, y_0, z_0) with $y_0 z_0 \neq 0$, is [JEE 2010, 3]

M.W

THANK - YOU 😊